Straight exercises from course 4

Exercice1 : Your first state-feedback control

Let us consider the system defined by the following differential equations, model very similar to the prey-predator system studied in the previous exercise. u is a food quantity. The implicite time unit is the month.

$$\left\{ \begin{array}{lcl} \dot{x}_1 & = & -x_1 + x_1 x_2 \\ \dot{x}_2 & = & x_2 - x_1 x_2 + u \end{array} \right. \tag{1}$$

Linearization

1.1 For $u_e = 2$, determine the equilibrium points.

1.2 Determine the linearized model around these equilibrium points

1.3 Study their stability. Is the linearized model is controllable ?

About the controllable linearized model, pole placement

Around the adequate equilibrium point, the aim is to determine a controller which, despite the continuous arrival of predators, manages to maintain population equilibrium by avoiding causing too many variations, with time response of less than 0.5 months. In control science language, this can be translated as not causing any overshoot. The proposed control is as follows

$$\tilde{u} = -K\tilde{x}$$

1.4 Propose the expected eigenvalues of the closed-loop system

1.5 Determine K that satisfies the specification

1.6 Using the joint simulink file, implement this control, and propose a discussion about the behavior you get.

About the controllable linearized model, LQ control

We now want to implement the same kind of control $\tilde{u} = -K_{LQ}\tilde{x}$, in which K_{LQ} results from an optimal LQ control. The criterion to optimize is

$$\int_0^{+\infty} \left(\tilde{x}^{\mathrm{T}}(\tau) Q \tilde{x}(\tau) + \tilde{u}^2(\tau)\right) \mathrm{d}\tau$$

The objective is to understand the impact of the weighting matrix $Q = \begin{pmatrix} q_1 & 0 \\ 0 & q_2 \end{pmatrix}$

1.7 Check the conditions of existence of a LQ control

1.8 Using the joint simulink file, implement the control for different choices of q_1 and q_2

1.9 Propose a summary of the experiments carried out

Implementation on the nonlinear model

1.10 What is the expression of the control of the nonlinear system ?

1.11 Using the second joint simulink file, implement the control.

1.12 As in the previous sections, discuss for different control laws the behavior of the nonlinear model.

Exercice2 : From transfer function to state-feedback control

Let us consider a system described by the following transfer function :

$$\frac{Y}{U}(s) = \frac{s+1}{s(s+10)^2} \tag{2}$$

The specifications are :
— overshoot : 5%
— time response : 2 seconds
— No steady-state error

We suppose that the state is accessible, no matter the choice of the state-space representation

2.1 Determine a state-feedback control of the following form that ensures the specifications

$$u = -Kx + ly^r$$

2.2 Complete the joint matlab file and implement the solution in simulink.